ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **The Human Eye and the Colourful World** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** optical instrument	It forms images like a camera.
2	**(d)** myopic	Near-sighted.
3	**(a)** two triangular bases and three rectangular lateral surfaces	Glass prism.
4	**(b)** cornea	Cornea.
5	**(a)** hypermetropic	Far-sighted.
6	**(d)** spectrum	Light spectrum.
7	**(d)** iris	Iris.
8	**(c)** concave lenses	Diverging.
9	**(d)** Violet	Violet.
10	**(c)** pupil	Light enters here.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Retina (2).	2
12	Myopia (near-sightedness) (2).	2
13	Most: violet (1); Least: red (1).	1+1
14	Iris (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Iris and pupil control light (1); Cornea and lens focus the light (1); Retina receives the image (1).	1+1+1
16	Weakening of ciliary muscles (1); Reduced flexibility of the lens (1); Near point recedes with age (1).	1+1+1
17	No atmosphere in space (1); No scattering of sunlight (1); So the sky appears black (1).	1+1+1
	OR: Myopia: far objects blurred; hypermetropia: near objects blurred (1); Myopia image in front of retina; hypermetropia image behind (1); Myopia: concave lens; hypermetropia: convex lens (1).	1+1+1
18	Splitting of white light into colours (1); Violet deviates the most (1); Red deviates the least (1).	1+1+1
	OR: Water droplets are big (1); They scatter all wavelengths equally (1); The scattered light is white (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Cornea (1); Iris and pupil (1); Lens and ciliary muscles (1); Retina (1); Optic nerve (1).	1+1+1+1+1
	OR: Sunlight falls on raindrops (1); Refraction at the first surface (1); Dispersion into colours (1); Internal reflection (1); Second refraction and colours reach the eye (1).	1+1+1+1+1
20	Light enters through cornea and pupil (1); Lens focuses it on the retina (1); Image is real and inverted (1); Retina sends signals through the optic nerve (1); Brain interprets them erect (1).	1+1+1+1+1
	OR: Atmosphere has layers of different density (1); Light bends continuously (1); Stars appear to twinkle (1); Sun seen before sunrise (1); Sun seen after sunset (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that a myopic eye is corrected with a convex lens, or hypermetropia with a concave lens; myopia needs concave and hypermetropia needs convex.
- Believing that most refraction in the eye occurs at the eye lens; it occurs at the outer surface of the cornea, and the lens only fine-tunes the focal length.
- Writing that the image on the retina is erect; it is real and inverted and the brain interprets it as erect.
- Stating that red light is scattered the most; it is scattered the least, and blue light is scattered the most.
- Forgetting the negative sign for a myopia lens, or using f in centimetres to find power; $P = 1/f$ with f in metres.

- Saying that planets twinkle like stars; stars twinkle due to atmospheric refraction of a point source, whereas planets appear as extended sources.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **The Human Eye and the Colourful World** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** convex lenses	Converging.
2	**(a)** Red	Red.
3	**(c)** a transparent, flexible material	Fibrous jelly-like.
4	**(c)** Accommodation	Accommodation is a normal function.
5	**(a)** seven colours	VIBGYOR.
6	**(c)** optic nerve	From the retina to the brain.
7	**(c)** surgery	Lens is replaced.
8	**(a)** colloidal solutions	Particles scatter light.
9	**(d)** rods and cones	Photoreceptors.
10	**(a)** negative	f is negative.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Hypermetropia (far-sightedness) (2).	2
12	Scattering of light by colloidal particles (2).	2
13	The minimum distance of distinct vision (1); About 25 cm (1).	1+1
14	Weakening of near vision with age (1); Due to weak ciliary muscles (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Adjustment of the focal length of the eye lens (1); Ciliary muscles contract (1); Lens becomes thicker (1).	1+1+1
16	Violet, indigo, blue (1); Green, yellow (1); Orange, red (1).	1+1+1
17	Red has the longest wavelength (1); Scattered the least by fog (1); Visible from far (1).	1+1+1
	OR: P = 1/f (1); P = 1/(-0.5) = -2 D (1); It is a concave lens (1).	1+1+1
18	Bending of light by the atmosphere (1); Twinkling of stars (1); Advance sunrise (1).	1+1+1
	OR: Deflection of light by particles in all directions (1); Blue sky (1); Red sun at sunrise (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Cannot see far objects clearly (1); Image is formed in front of the retina (1); Cause: lens too curved or eyeball elongated (1); Corrected by a concave lens (1); Far point is nearer than infinity (1).	1+1+1+1+1
	OR: Atmospheric particles scatter light (1); Blue light is scattered more (1); Sky is blue (1); Sunrise rays travel a long path (1); Red is the least scattered (1).	1+1+1+1+1
20	Cannot see near objects clearly (1); Image forms behind the retina (1); Cause: eyeball too short or focal length too large (1); Corrected by a convex lens (1); Near point is farther than 25 cm (1).	1+1+1+1+1
	OR: Scattering of light by colloidal particles (1); Path of the beam is visible (1); Example: sunlight in a dense forest (1); Example: a beam of light in a dusty room (1); Particles scatter in all directions (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that a myopic eye is corrected with a convex lens, or hypermetropia with a concave lens; myopia needs concave and hypermetropia needs convex.
- Believing that most refraction in the eye occurs at the eye lens; it occurs at the outer surface of the cornea, and the lens only fine-tunes the focal length.
- Writing that the image on the retina is erect; it is real and inverted and the brain interprets it as erect.
- Stating that red light is scattered the most; it is scattered the least, and blue light is scattered the most.
- Forgetting the negative sign for a myopia lens, or using f in centimetres to find power; $P = 1/f$ with f in metres.

- Saying that planets twinkle like stars; stars twinkle due to atmospheric refraction of a point source, whereas planets appear as extended sources.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **The Human Eye and the Colourful World** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** scattering	Blue light scatters more.
2	**(b)** becomes larger	Allows more light in.
3	**(d)** dioptre	1 D = 1 m^-1.
4	**(a)** scattering	Blue is scattered away.
5	**(a)** becomes smaller	Protects the retina.
6	**(c)** presbyopia	Upper part concave, lower part convex.
7	**(d)** atmospheric refraction	Changing air density.
8	**(a)** convex lens	Converging lens.
9	**(d)** food rich in vitamin A	Vitamin A is needed.
10	**(a)** rain, with the Sun behind us	Droplets in air disperse sunlight.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Blue light is scattered the most by air molecules (2).	2
12	The farthest point seen clearly (1); Infinity (1).	1+1
13	Band of colours obtained by dispersion of light (1); VIBGYOR (1).	1+1
14	Atmospheric refraction (1); Changing density of air (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	f = 1/P = 0.5 m = 50 cm (1); Positive power means a convex lens (1); Defect: hypermetropia (1).	1+1+1
16	Atmosphere bends sunlight (1); Sun appears above the horizon before actually rising (1); About 2 minutes earlier (1).	1+1+1
17	Red light is bent (1); No further colours appear (1); Red is monochromatic (1).	1+1+1
	OR: Cannot see distant objects (1); Myopia (1); Concave lens (1).	1+1+1
18	A natural spectrum seen in the sky (1); Raindrops act as tiny prisms (1); Refraction, dispersion and internal reflection (1).	1+1+1
	OR: Planets are nearby extended sources (1); Many point sources average out (1); Variations cancel (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Image must be formed at 50 cm (1); u = infinity, v = -50 cm (1); f = -50 cm (1); P = 1/f = -2 D (1); Concave lens (1).	1+1+1+1+1
	OR: Starlight passes through layers of changing air density (1); Its path bends and keeps changing (1); Stars are point sources, so brightness flickers (1); Planets are extended sources (1); Variations cancel out (1).	1+1+1+1+1
20	White light bends on entering the prism (1); Each colour bends by a different angle (1); Violet bends the most (1); Red bends the least (1); VIBGYOR spectrum (1).	1+1+1+1+1
	OR: Lenses with two parts (1); Upper part is concave for distance (1); Lower part is convex for near work (1); Correct presbyopia (1); When a person has both myopia and hypermetropia (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that a myopic eye is corrected with a convex lens, or hypermetropia with a concave lens; myopia needs concave and hypermetropia needs convex.
- Believing that most refraction in the eye occurs at the eye lens; it occurs at the outer surface of the cornea, and the lens only fine-tunes the focal length.
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- Saying that planets twinkle like stars; stars twinkle due to atmospheric refraction of a point source, whereas planets appear as extended sources.

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